



Luca Sportelli

DOCTORAL RESEARCHER
SPACECRAFT AUTONOMY

CONTACT

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PROFILES

Google Scholar
GitHub
LinkedIn

RESEARCH FOCUS

AI for GNC
Rendezvous and proximity
operations
Vision-based navigation
Small-body navigation
Interpretable autonomy

CORE TOOLS

Python, C/C++, MAT-
LAB/Simulink, Simscape
STK, ESATAN,
Capella/MBSE, DOORS
L^AT_EX

CERTIFICATIONS

IELTS Academic 8.0
L 9.0 · R 8.5 · W 6.5 · S 7.0
IPMA Introductory Certificate
GRE, Mar 2022
Ancient Latin Language Certifi-
cation, May 2017

Academic CV

01 / 01

RESEARCH / ENGINEERING / EXPLORATION

AI for spacecraft GNC and autonomy

Doctoral researcher developing AI-enabled methods for spacecraft guidance, navigation and control. Research focuses on temporal vision-based navigation, rendezvous and proximity operations, AI-augmented model predictive control, and interpretable spacecraft autonomy.

RESEARCH & EXPERIENCE

Doctoral Researcher · XDLab, ERAU

Aug 2025–Present

Developing temporal vision-based navigation and AI-augmented model predictive control for autonomous spacecraft operations. Advisor: Prof. Di Wu.

Thermal Control System Team Leader · PoliSpace 6S

Apr 2022–Mar 2024

Led thermal design and analysis for the student-developed 1U CubeSat using ESATAN and MATLAB/Simulink, including thermal strategy definition, subsystem interfaces and team coordination. Team selected for ESA Fly Your Satellite! Design Booster.

Analog Astronaut · Asclepios IV / EuroMoonMars

2022–2024

Selected worldwide as one of eight Selene Crew analog astronauts for Asclepios IV through a six-month ESA-like selection process, followed by one year of mission training under mentors including former ESA astronaut Claude Nicollier and former ESA expert Prof. Marc Toussaint; also served in CapCom activities. Separately participated in the EuroMoonMars EMMPOL 13 analog mission in Poland as Data Officer within an international crew during a two-week isolation campaign.

EDUCATION

PhD Aerospace Eng. · Embry-Riddle Aeronautical University

Aug 2025–Present

Research in autonomous spacecraft navigation and control. Advisor: Prof. Di Wu.

MSc Space Engineering · Politecnico di Milano

Sep 2022–Dec 2024

110/110 cum laude; thesis on particle propagation and imaging simulation in the Didymos binary system. Fabrizio Schiavi Scholarship and DSU Scholarships, 2022–2023.

Alta Scuola Politecnica · Politecnico di Milano

Feb 2023–Aug 2024

Highly selective honours programme completed alongside the MSc, combining additional multidisciplinary coursework with a team-based interdisciplinary project. Served as Team Controller for the project team.

BSc Aerospace Engineering · Politecnico di Milano

Sep 2019–Jul 2022

108/110; DSU Scholarships, 2019–2021.

SELECTED PUBLICATIONS & THESIS

AI-Augmented Model Predictive Control for Safe and Adaptive Rendezvous and Proximity Operations

L. Sportelli, T. Barr, C. Kilic, D. Wu · arXiv:2607.16630 · 2026

Design Principles for Explainable and Adaptive AI in Space Differential Games

T. Barr et al. (incl. L. Sportelli) · AAS/AIAA Astrodynamics · 2026

Modeling Particle Propagation in the Didymos Binary System for Imaging Simulation and Particle Rendering

L. Sportelli · MSc thesis · Politecnico di Milano · 2024

Structural and Thermal Model Testing Campaign of a 1U CubeSat

F. Botti et al. (incl. L. Sportelli) · 75th IAC · 2024

Design and Architecture of Anubis: A Hopper to Sample and Study Mercury's Surface and Subsurface

A. Cagnoni et al. (incl. L. Sportelli) · 75th IAC · 2024

PROFESSIONAL STRENGTHS

Teamwork · Leadership · Quick Learning · Organisation · Drive · Quality Focus

Clickable links provide access to profiles, publications, reports and project documentation.